

Learning Networks*

Promit K. Chaudhuri[†] Jonghyun Park[‡] Sudipta Sarangi[§]
Hector Tzavellas[¶]

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Abstract

We study repeated games in which agents observe only local actions and do not know how information flows across others. This creates uncertainty about both opponents' behavior and the interaction structure that determines payoffs and observations. We consider a setting in which this structure is not commonly known and ask whether it can be learned from repeated play. Agents update beliefs based on their observations and choose actions that are optimal given their current beliefs. We show that behavior converges in finite time to the complete-information Nash equilibrium of the realized stage game, even without full identification of the interaction structure. We provide conditions under which the structure is identified and characterize how the speed of learning depends on network topology. A linear-quadratic example shows that when equilibrium actions reflect walk profiles in the network, then they can generically reveal the underlying structure.

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[†]Department of Economics, Virginia Tech

[‡]Department of Economics, Virginia Tech

[§]Department of Economics, Virginia Tech

[¶]Department of Economics, Virginia Tech

1 Introduction

Many economic environments are characterized by decentralized interactions in which agents observe only local outcomes and have limited knowledge of how information is distributed across others. In social and economic networks, individuals typically observe the behavior of their neighbors but do not know who influences whom beyond their immediate peers. In organizations, firms or workers may observe the actions of close collaborators without knowing the broader structure through which information and incentives propagate. In such settings, agents face a fundamental problem. They must form expectations about others based not only on what they observe, but also on an incomplete understanding of what others observe. This raises a natural question. How do agents learn and behave in dynamic environments where the structure of interactions, and therefore the flow of information, is uncertain?

Existing work on Bayesian learning in repeated games abstracts away from this question, broadly maintaining common knowledge of the interaction structure. For instance, in models with perfect monitoring, all players observe the actions of all others, and learning concerns only opponents' behavior within a known environment (Jordan, 1991; 1995; Kalai and Lehrer, 1993a; b). Similarly, with imperfect monitoring, while players observe only signals of underlying actions, the signal structure itself, whether given by a general distribution or a fixed monitoring graph, is typically assumed to be commonly known (Renault and Tomala, 1998; 2004). Across these approaches, the monitoring structure is either global and commonly known or, when uncertain, treated as a fixed informational primitive. None addresses whether agents can learn the structure of interactions from play in settings where flow of information is itself the object of Bayesian inference.

We address this gap by reframing the source of uncertainty. Our central observation is that uncertainty about the monitoring structure can be equivalently represented as uncertainty about the interaction structure. Since monitoring determines who observes whom, not knowing the monitoring structure is equivalent to not knowing the network of interactions that governs both payoffs and observations. This equivalence links repeated games with imperfect monitoring to network games with incomplete information and allows the unknown interaction structure to be treated as a state that agents learn through repeated play.

We study a repeated game in which interactions are governed by an underlying network that determines both payoffs and observations. The network is fixed but unknown, and agents are endowed with a common prior over a set of possible networks. Each agent observes only her local neighborhood and receives signals given by the actions of those with whom she interacts

directly. In this way, the monitoring structure is determined by the interaction network and is not commonly known. Agents maintain Bayesian beliefs over the set of possible networks and update these beliefs based on the sequence of locally observed actions. This formulation captures precisely the environment described above. Agents do not know who observes whom or how information propagates, and must infer these from the actions of neighbors over time. At the same time, the model preserves Bayesian inference, allowing us to study learning and behavior in a setting where the information structure itself is uncertain.

Our first result characterizes the convergence of equilibrium behavior. We focus on myopic agents who update beliefs about the interaction structure over time based on the sequence of actions they observe locally. Agents choose actions that are optimal given their current beliefs. Since observations accumulate over time, the set of networks consistent with a player’s observations shrinks monotonically and eventually stabilizes. We show that once this stabilization occurs, equilibrium play coincides with the complete-information Nash equilibrium of the stage game played on the realized interaction structure.

A key feature of this result is that convergence does not require full identification of the network. Learning transpires on the interaction structure rather than on opponents’ behavior, and once beliefs stabilize, payoffs being dependent only on actions of neighbors ensure that any residual uncertainty about the network is payoff-irrelevant. This departs from standard Bayesian learning environments in a fundamental way. There, players form beliefs over opponents’ strategies under global monitoring, and convergence to Nash play arises asymptotically because predictive distributions converge to the distribution induced by actual play. Here, the common prior over networks already aligns beliefs about behavior conditional on the structure, so inference operates on the structural primitive under local observation. Partial structural learning is therefore sufficient to guarantee action convergence.

While behavioral convergence does not rely on full identification of the interaction structure, perfect learning is nonetheless possible. Specifically, it requires equilibrium behavior to be sufficiently informative to distinguish among interaction structures that remain consistent with a player’s observations. Our second result characterizes conditions under which this holds. We show that if the complete-information equilibrium correspondence separates interaction structures through the action of a player or one of her observed neighbors, then the realized structure is identified once learning stabilizes. In this case, equilibrium behavior eliminates all remaining observationally indistinguishable interaction structures, so that agents not only behave as if the structure were known, but also learn it exactly.

More generally, this result highlights that identification is driven by how equilibrium behavior maps interaction structures into actions. If any two interaction structures that are

observationally indistinguishable to a player (or to one of her observed neighbors), induce different equilibrium actions, then the player will identify the network once learning stabilizes. Therefore, while local observations may suffice for correct long-run behavior, full learning requires that these observations encode the structure in a way that separates all observationally equivalent networks.

We concretely illustrate this mechanism in a linear-quadratic network game, where complete information (stage-game) equilibrium actions admit a closed-form expression in terms of Katz-Bonacich centrality. These actions summarize not only agents direct connections, but the entire pattern of walks of all lengths they have in the network. This representation makes the mapping from interaction structures to actions explicit. We show that if two networks differ in the number of walks of any given length reaching a player, then the corresponding equilibrium actions differ for almost all parameter values. Thus, differences in higher-order connectivity are reflected in behavior and break the observational equivalence of networks that cannot be distinguished through local observations alone. As a consequence, if a player, or one of her observed neighbors, faces a set of observationally indistinguishable networks that differ in the number of walks of some length, then equilibrium actions will separate these networks once learning stabilizes, leading to identification.

Lastly, we study the speed of learning. Since agents refine their beliefs by eliminating interaction structures that are inconsistent with observed behavior, learning proceeds through a sequence of finite reductions and therefore concludes in finite time. We first establish a coarse bound that depends only on the size of the set of possible interaction structures. This bound reflects the fact that each observation can rule out only a limited number of candidate structures. We then show that sharper bounds can be obtained by exploiting the local nature of observation. Because agents observe only their neighbors, information about distant parts of the network must propagate through a sequence of interactions. As a result, the time required for learning is governed by the topology of the network, and in particular by the distance over which information must travel. This highlights a key feature of the environment. Learning is not only contingent on the accumulation of observations, but also depends on how information flows through the interaction structure.

Related literature

A natural benchmark for our paper is the Bayesian learning literature, which primarily focuses on the issue of convergence to Nash equilibrium under repeated play and perfect monitoring. [Jordan \(1991\)](#) shows that myopic Bayesian play in finite normal-form games

converges almost surely to the Nash set, while [Jordan \(1995\)](#) extends this to forward-looking players, showing that play asymptotically converges to the Nash equilibrium of the repeated game. [Kalai and Lehrer \(1993a\)](#) establish the same conclusion when each player’s prior assigns positive probability to strategies opponents employ during the game, while [Kalai and Lehrer \(1993b\)](#) relax this informational restriction by showing that beliefs need only correctly predict opponents’ actions on the equilibrium path. We depart from this literature in two ways. First, the object being learned in our game is the interaction network rather than opponents’ strategies. Second, in our model, convergence occurs in finite time.

A related strand studies repeated games with imperfect monitoring, where players do not directly observe opponents’ actions but instead receive noisy signals drawn from a commonly known distribution. Folk theorems in this literature characterize sustainable equilibrium payoffs under public monitoring ([Fudenberg et al., 1994](#)), private monitoring with public communication ([Compte, 1998](#); [Kandori and Matsushima, 1998](#)), and private monitoring without communication ([Hörner and Olszewski, 2006](#); [Sugaya, 2022](#)). Our environment changes the informational primitive: signals are perfect but the structure that determines them is unknown. Correspondingly, the focus of our analysis is on how players learn this structure from local observations and their resulting behaviors, rather than the set of sustainable equilibrium payoffs.

When the monitoring structure itself is uncertain, the literature typically weakens the equilibrium concept. [Fudenberg and Levine \(1993\)](#) introduce self-confirming equilibrium, in which beliefs about opponents’ strategies need only be consistent with observations along the equilibrium path. [Battigalli et al. \(2023\)](#) extend this notion to network games in which players observe only their realized payoffs, and characterize long-run play. [Lipnowski and Sadler \(2019\)](#) introduce peer-confirming equilibrium, in which a network encodes which players have correct conjectures about each other’s strategies. We share the view that networks naturally restrict what each player can know about others, but retain a common prior and Bayesian updating throughout. The network in our setting is not an exogenous informational restriction but the latent state that players learn from local observations over time.

A separate literature studies repeated games in which monitoring is local rather than global, restricting who observes or communicates with whom. This monitoring structure is nonetheless taken to be common knowledge ([Ben-Porath and Kahneman, 1996](#); [Laclau, 2014](#); [Renault and Tomala, 1998](#)). Closer to our environment is [Nava and Piccione \(2014\)](#) who consider a monitoring structure determined by a fixed network in which each player privately observes only her own neighborhood, and [Wolitzky \(2013\)](#) who specifies a monitoring network drawn each period from a commonly known distribution. In both cases, they ask

which beliefs over the monitoring structure and which features of that structure will sustain long run cooperation. Although our analysis also concerns how beliefs over the monitoring structure shape long run play, our focus is on whether that structure can itself be learned. We study how repeated local observation refines beliefs about the monitoring structure over time and shapes the resulting equilibrium behavior.

Our payoff environment belongs to the class of network games with local spillovers, in which a player’s payoff depends only on her own action and the actions of their neighbors (Ballester et al., 2006; Goyal and Joshi, 2003; 2006). A separate body of work introduces incomplete information into such games in static settings. Galeotti et al. (2010) study behavior games in which agents only know their degree, while Martí and Zenou (2015) analyze incomplete information about payoff-relevant parameters and characterize Bayesian-Nash equilibria. Closest to our informational environment is Chaudhuri et al. (2026), who study static network games under network uncertainty when agents know only their local connections. We extend their setting to a dynamic environment where the interaction structure is initially unknown, but locally observed actions accumulate information over time resolving the underlying network uncertainty.

Our paper is also related to the literature on Bayesian learning in networks. Renault and Tomala (2004) and Li and Tan (2020) study how agents learn an unknown state under graph-based monitoring. Gale and Kariv (2003), Mueller-Frank (2013), and Huang et al. (2024) analyze how agents infer information from neighbors’ actions, while Dasaratha and He (2026) quantify how network structure shapes the efficiency of signal aggregation. We differ on the object of inference: the latent state in our setting is the network itself, and informative signals take the form of endogenous equilibrium actions that reflect agents positions in interaction structure. Convergence therefore need not take the form of consensus, as in the imitation principle of Mueller-Frank (2013); instead, the limiting outcome is the complete-information Nash equilibrium of the realized network game.

Finally, our paper is also complementary to the large literature on boundedly rational or non-Bayesian learning in networks (Acemoglu et al., 2010; Bala and Goyal, 1998; Dasaratha and He, 2020; Golub and Jackson, 2010). That literature typically asks whether decentralized updating aggregates dispersed information. We instead maintain full Bayesian rationality and study a different friction: players observe only a local subset of equilibrium actions and use those observations to learn an initially unknown interaction structure. Our contribution is to characterize when this local inference process stabilizes, when it identifies the true network, and when the resulting behavior coincides with the complete-information Nash equilibrium of the realized network game.

The rest of the paper is structured as follows. Section 2 sets up the game. Section 3 presents the convergence and network identification results. Section 4 illustrates the results in a linear quadratic game. Section 5 characterizes the speed of learning. Section 6 concludes.

2 A Repeated Game under Network Uncertainty

We study a repeated game of strategic interaction in a finite population. In each period, a player's stage payoff depends on the actions of a subset of opponents with whom she interacts, and at the end of the period she observes only the actions of those same opponents. The interaction structure or network is drawn once at the beginning of the game and remains fixed thereafter. It is not common knowledge: each player observes only her own local neighborhood and holds beliefs about the rest of the structure. Over time, players update these beliefs using signals generated by repeated play.

A distinctive feature of the model is that the unknown state is the interaction structure itself, and the signals from which players learn are the actions they observe locally. In this sense, the model combines strategic interaction with decentralized monitoring, where players observe only a restricted subset of actions rather than the full action profile. This stands in contrast to public-monitoring environments, in which all players observe either a common public signal or the entire profile of actions after each period. Here, learning is necessarily decentralized, since each player's information is determined by her position in the interaction structure and by the observations generated along the path of play.

Primitives

There is a finite set of players $N = \{1, \dots, n\}$. For each $i \in N$, let $A_i \subseteq \mathbb{R}_+$ be a nonempty action set. For any subset $S \subseteq N$, let $A_S := \prod_{i \in S} A_i$, and write $A := A_N$.

Time is discrete and indexed by $\tau = 0, 1, 2, \dots$. In each period $\tau \geq 1$, players choose actions simultaneously. Let $a_i^\tau \in A_i$ denote player i 's action in period τ , and let $a^\tau = (a_i^\tau)_{i \in N} \in A$ denote the resulting action profile.

Interaction Structure

An *interaction structure* is a profile $\mathbf{g} = (N_i(\mathbf{g}))_{i \in N}$, where for each $i \in N$, $N_i(\mathbf{g}) \subseteq N \setminus \{i\}$. The set $N_i(\mathbf{g})$ represents the players with whom i interacts directly. The interaction structure determines both whose actions affect player i 's payoff and whose actions player i

observes at the end of each period.¹ This formulation captures environments in which the same network of local relationships determine both strategic spillovers and the information available through repeated observation.

Let $\mathcal{G}_n$ be a finite set of feasible interaction structures. At $\tau = 0$, nature draws $\mathbf{g} \in \mathcal{G}_n$ according to a commonly known prior $p \in \Delta(\mathcal{G}_n)$ with full support, and the realized structure remains fixed thereafter.

After $\mathbf{g}$ is realized, player i observes only her local interaction set $N_i(\mathbf{g})$. Her initial belief at $\tau = 1$ is therefore the prior p conditioned on the set of structures consistent with $N_i(\mathbf{g})$, that is, $p(\cdot \mid N_i(\mathbf{g}))$.

This informational environment is closely related to [Chaudhuri et al. \(2026\)](#), who study static network games under network uncertainty in which agents observe their local connections and update their beliefs about the rest of the network using bayes rule. We adopt the same informational premise, but embed it in a repeated game with local monitoring. The focus here is therefore not only on static equilibrium under network uncertainty, but on how decentralized observations generated by repeated play refine beliefs over time and more importantly whether they identify the realized interaction structure.

Stage Payoffs

For each player i and interaction structure $\mathbf{g} \in \mathcal{G}_n$, stage payoffs are described by a *local spillover function*

$$v_i^{\mathbf{g}} : A_i \times A_{N_i(\mathbf{g})} \rightarrow \mathbb{R},$$

with $v_i^{\mathbf{g}} = v_i^{\mathbf{g}'}$ whenever $N_i(\mathbf{g}) = N_i(\mathbf{g}')$. Given an action profile $a \in A$, player i 's payoff is

$$u_i(a, \mathbf{g}) = v_i^{\mathbf{g}}(a_i, a_{N_i(\mathbf{g})}).$$

Thus, player i 's payoff depends only on her own action and the actions of her neighbors. In particular,

$$u_i(a, \mathbf{g}) = u_i(a, \mathbf{g}') \quad \text{whenever } N_i(\mathbf{g}) = N_i(\mathbf{g}').$$

This restriction implies that player i 's payoff is fully determined by her neighborhood $N_i(\mathbf{g})$. The formulation nests standard linear-quadratic network games, where an agent's payoff depends on her own action and the actions of her neighbors ([Ballester et al., 2006](#)), and more

¹More generally, one could distinguish between an influence set $N_i^u(\mathbf{g})$, which determines payoff relevance, and an observation set $N_i^z(\mathbf{g})$, which determines observed signals. We focus on the benchmark case in which these sets coincide.

generally captures local-spillover environments in which payoff-relevant interactions are mediated through direct links (Bramoullé et al., 2014). In R&D collaboration networks, for example, a firm's research payoff may depend on the R&D effort, cost reductions, or innovation outcomes of its collaborators, as in models of collaboration networks in oligopoly and R&D network formation (Billand et al., 2019; 2026; Goyal and Joshi, 2003; 2006). Similar local payoff externalities arise in network games with peer effects, workplace interactions, and local public-good environments, where an agent's payoff-relevant environment is determined by the behavior of those with whom she directly interacts (Calvó-Armengol et al., 2009; Jackson and Zenou, 2015; Lindquist et al., 2026).

We impose the following standard regularity conditions on stage payoffs.

Assumption 1. *For every player $i \in N$ and every interaction structure $\mathbf{g} \in \mathcal{G}_n$: (i) A_i is compact and convex, (ii) $v_i^{\mathbf{g}}$ is continuous, and (iii) for every $a_{N_i(\mathbf{g})} \in A_{N_i(\mathbf{g})}$, the map $a_i \mapsto v_i^{\mathbf{g}}(a_i, a_{N_i(\mathbf{g})})$ is concave on A_i .*

Local Observation and Private Histories

At the end of each period, player i does not observe the full action profile. Instead, she observes only her own action and the actions of the players in her neighborhood $N_i(\mathbf{g})$. This relates to Bayesian learning in social networks, where agents observe their neighbors' actions to learn an underlying state (Gale and Kariv, 2003; Huang et al., 2024; Mueller-Frank, 2013). There the network channels information about an exogenous state. Here it is the unknown state itself: observed actions are informative because the structure that generates them is the one that determines payoffs.

Formally, player i 's observation is given by a signal function

$$z_i : A \times \mathcal{G}_n \rightarrow Z_i, \quad z_i(a, \mathbf{g}) = (a_i, (a_j)_{j \in N_i(\mathbf{g})}),$$

where $Z_i := \bigcup_{S \subseteq N \setminus \{i\}} (A_i \times A_S)$. If a^τ is the action profile in period τ , the realized signal is $z_i^\tau = z_i(a^\tau, \mathbf{g})$.

Player i 's private history at the beginning of period $\tau \geq 1$ is

$$h_i^\tau = \begin{cases} N_i(\mathbf{g}) & \text{if } \tau = 1, \\ (N_i(\mathbf{g}), z_i^1, \dots, z_i^{\tau-1}) & \text{if } \tau \geq 2. \end{cases}$$

Thus, a private history consists of the observed neighborhood together with the sequence of locally observed signals. In particular, at $\tau = 1$, the history reduces to the observed neighborhood.

The set of all possible private histories for player i at period τ is

$$\mathcal{H}_i^\tau = \bigcup_{S \subseteq N \setminus \{i\}} \{S\} \times (A_i \times A_S)^{\tau-1}.$$

Under a realized interaction structure $\mathbf{g}$ and a profile of past play, player i 's realized history satisfies $h_i^\tau \in \mathcal{H}_i^\tau$.

Strategies and Induced Play

A pure strategy for player i is a sequence $\sigma_i = (\sigma_i^\tau)_{\tau \geq 1}$, where

$$\sigma_i^\tau : \mathcal{H}_i^\tau \rightarrow A_i$$

assigns an action to each period- τ private history. A pure-strategy profile is denoted by $\sigma = (\sigma_i)_{i \in N}$.

Given an interaction structure $\mathbf{g} \in \mathcal{G}_n$ and a strategy profile σ , play is defined recursively. For each player i , let

$$h_i^1(\mathbf{g}, \sigma) := N_i(\mathbf{g}).$$

For each period $\tau \geq 1$, define

$$a_i^\tau(\mathbf{g}, \sigma) = \sigma_i^\tau(h_i^\tau(\mathbf{g}, \sigma)), \quad a^\tau(\mathbf{g}, \sigma) = (a_k^\tau(\mathbf{g}, \sigma))_{k \in N},$$

and

$$z_i^\tau(\mathbf{g}, \sigma) = z_i(a^\tau(\mathbf{g}, \sigma), \mathbf{g}).$$

The private history then evolves according to

$$h_i^{\tau+1}(\mathbf{g}, \sigma) = (N_i(\mathbf{g}), z_i^1(\mathbf{g}, \sigma), \dots, z_i^\tau(\mathbf{g}, \sigma)).$$

Thus, for each $\mathbf{g}$ and σ , the model induces a unique sequence

$$\{(h^\tau(\mathbf{g}, \sigma), a^\tau(\mathbf{g}, \sigma), z^\tau(\mathbf{g}, \sigma))\}_{\tau \geq 1}.$$

Distinct interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$ may generate identical private histories for player i under σ , and are therefore indistinguishable from her perspective. This observation motivates the following definition.

Definition 1. Fix a strategy profile σ , a player i , and a period $\tau \geq 1$. Two interaction

structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$ are observationally equivalent for player i at period τ under σ , written $\mathbf{g} \sim_{i,\tau} \mathbf{g}'$, iff $h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau(\mathbf{g}', \sigma)$. We denote by $[\mathbf{g}]_{i,\tau}$ the corresponding equivalence class.²

Beliefs

Fix a player i and a period $\tau \geq 1$. Since player i 's private history at the beginning of period τ is determined by the realized interaction structure together with play in periods $1, \dots, \tau - 1$, the prior p and the strategy profile σ induce a probability distribution over $\mathcal{H}_i^\tau$. For any history $h_i^\tau \in \mathcal{H}_i^\tau$, define

$$P_{i,\sigma}(h_i^\tau) := \sum_{\mathbf{g} \in \mathcal{G}_n} p(\mathbf{g}) \mathbb{I}\{h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau\}.$$

Thus, $P_{i,\sigma}(h_i^\tau)$ is the ex ante probability that player i 's realized private history at the beginning of period τ is h_i^τ .

The strategy profile σ therefore induces, for each player i and each period τ , a posterior belief over interaction structures at every history in $\mathcal{H}_i^\tau$. Let the induced belief system be denoted by

$$\mu_\sigma = (\mu_{i,\sigma}^\tau)_{i \in N, \tau \geq 1},$$

where

$$\mu_{i,\sigma}^\tau(\cdot \mid h_i^\tau) \in \Delta(\mathcal{G}_n)$$

denotes player i 's posterior at history $h_i^\tau \in \mathcal{H}_i^\tau$. At every history such that $P_{i,\sigma}(h_i^\tau) > 0$, these beliefs are obtained by Bayes' rule:

$$\mu_{i,\sigma}^\tau(\mathbf{g} \mid h_i^\tau) = \frac{p(\mathbf{g}) \mathbb{I}\{h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau\}}{P_{i,\sigma}(h_i^\tau)}, \quad \text{for all } \mathbf{g} \in \mathcal{G}_n.$$

At histories $h_i^\tau \in \mathcal{H}_i^\tau$ such that $P_{i,\sigma}(h_i^\tau) = 0$, beliefs may be specified arbitrarily.

Define the path of play of player i under σ by

$$\mathcal{P}_i(\sigma) := \bigcup_{\tau \geq 1} \{h_i^\tau \in \mathcal{H}_i^\tau : P_{i,\sigma}(h_i^\tau) > 0\},$$

and define

$$\mathcal{P}(\sigma) := \bigcup_{i \in N} \mathcal{P}_i(\sigma)$$

as the path of play induced by σ .

²The dependence of the relation $\sim_{i,\tau}$ and $[\mathbf{g}]_{i,\tau}$ on σ is suppressed whenever the strategy profile is fixed.

Because the prior p has full support, for every $\mathbf{g} \in \mathcal{G}_n$, every player i , and every period τ , the induced history $h_i^\tau(\mathbf{g}, \sigma)$ satisfies

$$P_{i,\sigma}(h_i^\tau(\mathbf{g}, \sigma)) \geq p(\mathbf{g}) > 0.$$

Hence, every history generated by some interaction structure under σ lies on the path of play.

Equilibrium

Players are myopic in the sense that, in each period, they choose actions to maximize current expected payoff given their posterior beliefs.³ This corresponds to the dynamic counterpart of [Jordan \(1991\)](#), where the unknown state is the interaction structure $\mathbf{g}$ and beliefs are updated through locally observed signals z_i^τ . In contrast to models with forward-looking behavior (e.g., [Jordan \(1995\)](#) and [Kalai and Lehrer \(1993a\)](#)), repeated interaction here serves to generate information rather than intertemporal incentives.

Definition 2. *A strategy profile σ^* is a myopic Bayesian equilibrium if, for every player i , every period $\tau \geq 1$, and every history $h_i^\tau \in \mathcal{P}_i(\sigma^*)$,*

$$\sigma_i^{*\tau}(h_i^\tau) \in \arg \max_{a_i \in A_i} \sum_{\mathbf{g} \in \mathcal{G}_n} \mu_{i,\sigma^*}^\tau(\mathbf{g} \mid h_i^\tau) u_i(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g}),$$

where $h_{-i}^\tau(\mathbf{g}, \sigma^*) = (h_j^\tau(\mathbf{g}, \sigma^*))_{j \neq i}$.

The equilibrium condition constrains behavior only on the path of play $\mathcal{P}(\sigma^*)$. At histories $h_i^\tau \notin \mathcal{P}_i(\sigma^*)$, the strategy $\sigma_i^{*\tau}(h_i^\tau)$ is unconstrained, and two strategy profiles that agree on $\mathcal{P}(\sigma^*)$ are equivalent. Under [Assumption 1](#), a myopic Bayesian equilibrium exists.

The myopic equilibrium abstracts from intertemporal incentives and, in particular, from the use of actions to influence future beliefs or continuation play. As a result, equilibrium behavior reflects only players' beliefs about the underlying network, so that observed actions provide a direct mapping from beliefs to behavior. This feature is central to our analysis, as it isolates the informational content of play and underlies our identification and convergence results. In this sense, the equilibrium concept plays a role analogous to the best-response dynamics studied in the Bayesian learning literature ([Jordan, 1991; 1995; Kalai and Lehrer, 1993a; b](#)), adapted to an environment in which the monitoring structure is endogenous and unknown. The concept is also related to self-confirming equilibrium ([Battigalli et al., 2023](#);

³As discussed by [Golub and Sadler \(2016\)](#), myopic updating can be viewed as a tractable approximation to fully rational behavior under heavy discounting.

Fudenberg and Levine, 1993), with the distinction that beliefs here are fully disciplined by a common prior and Bayes' rule at all on-path histories.⁴

3 Convergence and Identification

This section studies behavior and information once learning stabilizes. Two distinct findings arise. First, agents may eventually behave *as if* they know the true interaction structure, in the sense that equilibrium play coincides with the complete-information Nash equilibrium under the realized structure. This need not imply that they identify the structure itself. Second, agents may or may not be able to *identify* the realized interaction structure. We now show that first property holds quite generally, whereas the second depends on whether equilibrium behavior is sufficiently informative to distinguish observationally equivalent structures.

Fix a myopic Bayesian equilibrium σ^* . Throughout this section, all histories, observational-equivalence classes, and induced actions are defined relative to σ^* . For each player $i \in N$, period $\tau \geq 1$, and interaction structure $\mathbf{g} \in \mathcal{G}_n$, let $[\mathbf{g}]_{i,\tau}$ denote the observational-equivalence class of $\mathbf{g}$ for player i at period τ , that is,

$$[\mathbf{g}]_{i,\tau} = \{\mathbf{g}' \in \mathcal{G}_n : h_i^\tau(\mathbf{g}', \sigma^*) = h_i^\tau(\mathbf{g}, \sigma^*)\}.$$

Because the prior p has full support, at any on-path history $h_i^\tau(\mathbf{g}, \sigma^*)$, the support of player i 's posterior coincides with $[\mathbf{g}]_{i,\tau}$.

Throughout, for each $\mathbf{g} \in \mathcal{G}_n$, let $NE(\mathbf{g}) \subseteq A$ denote the set of pure strategy Nash equilibria of the complete information stage game induced by $\mathbf{g}$ and let $a^c(\mathbf{g})$ be a generic element of $NE(\mathbf{g})$ with $a_i^c(\mathbf{g})$ denoting its i^{th} component. For each player $i \in N$, write $NE_i(\mathbf{g}) := \{a_i(\mathbf{g}) : a(\mathbf{g}) \in NE(\mathbf{g})\}$ for the projection of the complete-information Nash equilibrium set under $\mathbf{g}$ onto player i 's action space.

3.1 Convergence to Stage-Game Equilibrium

Observation in this environment is local: players observe only the actions of those in their interaction neighborhoods. We show that, once learning stabilizes, this local information

⁴With forward-looking players, actions could manipulate beliefs or sustain history-dependent play. This introduces signaling and punishment motives that confound their informational role. Section 3.1 shows our results extend under restrictions that remove such effects, so myopia serves as a tractable benchmark that isolates learning.

is nonetheless sufficient to align equilibrium behavior with the complete-information benchmark.

We begin by formalizing when learning stabilizes.

Definition 3. *Learning ends at period τ^* if, for all $\tau \geq \tau^*$,*

$$[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,\tau^*} \quad \text{for all } i \in N \text{ and } \mathbf{g} \in \mathcal{G}_n.$$

After τ^* , no player rules out further interaction structures on the basis of observed play. This does not require that players identify the true structure, only that the equilibrium path generates no additional eliminations. In particular, if $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$ for some $\tau \geq \tau^*$, then $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$, so $\mathbf{g}$ and $\mathbf{g}'$ generate the same next-period signal for player i .

Lemma 1. *For every myopic Bayesian equilibrium σ^* , there exists a finite period τ^* such that learning ends, i.e. for all $i \in N$ and $\mathbf{g} \in \mathcal{G}_n$,*

$$[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,\tau^*} \quad \forall \tau \geq \tau^*.$$

We now state the main result. It shows that once learning stabilizes, any residual uncertainty becomes payoff-irrelevant along the equilibrium path.

Theorem 1. *Let $\mathbf{g} \in \mathcal{G}_n$ be the realized interaction structure. If learning ends at τ^* under σ^* , then*

$$a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g}) \quad \forall \tau \geq \tau^*.$$

The logic is straightforward. Once learning ends, all interaction structures in $[\mathbf{g}]_{i,\tau^*}$ generate the same payoff-relevant local observations at the realized history. Residual uncertainty is therefore irrelevant for player i 's best-response problem.

Theorem 1 is the core behavioral implication. Even if players may fail to identify the realized interaction structure ($|[\mathbf{g}]_{i,\tau^*}|$ may exceed one) they behave as if it were common knowledge. Uncertainty about the network structure thus becomes behaviorally irrelevant in the long run.

All of this rests on myopia. Were players forward-looking, an action could be chosen for its effect on others' beliefs or on future play, and the correspondence between learning and complete-information behavior in Theorem 1 need no longer hold. We take this up at the end of the section, and find that it is restored under a mild condition.

The theorem departs from classical Bayesian learning results such as Kalai and Lehrer (1993a) and Jordan (1995), in two ways. First, we replace global monitoring with strictly local observability. Second, we obtain finite-time rather than asymptotic convergence.

The result also contrasts with the imitation principle in Mueller-Frank (2013). In that setting, agents observe neighbors' actions to learn about a common payoff-relevant state. Their payoffs, however, do not directly depend on those neighbors' actions. Since agents face the same decision problem, an agent who observes a neighbor can copy the neighbor's action, and this disciplines long-run disagreement. Our model differs because the realized network is itself payoff-relevant. Neighbors' actions enter payoffs through local strategic spillovers, so players in different network positions generally face different best-response problems. Therefore, learning need not produce imitation or consensus. Once the realized interaction structure is revealed, play converges to the complete-information Nash equilibrium of the realized network game, which may remain heterogeneous across players.

Consider, then, forward-looking play. Suppose σ^* is a perfect Bayesian equilibrium. Let τ^* denote the stabilization time from Lemma 1. Although $[g]_{i,\tau}$ remains constant for all $\tau \geq \tau^*$, a deviation may affect continuation values through its impact on beliefs or prescribed future play, even if it does not alter the observational-equivalence class along the equilibrium path. Residual uncertainty within $[g]_{i,\tau^*}$ may therefore be irrelevant for stage payoffs but relevant for continuation values, so equilibrium behavior need not coincide with the complete-information benchmark.

Remark. *The conclusion of Theorem 1 survives for forward-looking players once intertemporal feedback through continuation values is eliminated.*

Suppose that for all $\tau \geq \tau^*$, continuation strategies depend only on $[g]_{i,\tau^*}$ and unilateral deviations do not affect posterior beliefs or continuation play, then continuation payoffs are invariant to deviations. Each player's problem reduces to a static best-response problem given beliefs over $[g]_{i,\tau^*}$, and

$$a_i^\tau(g, \sigma^*) \in \text{BR}_i(a_{-i}^\tau(g, \sigma^*); g) \quad \text{for all } i,$$

so that $a^\tau(g, \sigma^*)$ coincides with the complete-information Nash equilibrium. This highlights that myopia is not essential per se, but serves to eliminate informational and strategic feedback through continuation values.

3.2 Perfect Learning

Theorem 1 shows that long-run behavior coincides with the complete-information benchmark. We now ask when a player identifies the realized interaction structure.

Definition 4. Fix $\mathbf{g} \in \mathcal{G}_n$. Learning is perfect for player i if

$$[\mathbf{g}]_{i,\tau^*} = \{\mathbf{g}\}.$$

Perfect learning is stronger than behavioral convergence. It requires that equilibrium behavior be sufficiently informative to eliminate all incorrect interaction structures, and not merely to pin down equilibrium actions. Thus, Theorem 1 implies that play converges to the correct action profile, but not necessarily that players identify the underlying structure.

Theorem 2. Fix $\mathbf{g} \in \mathcal{G}_n$ and suppose learning ends at τ^* under σ^* . Fix $i \in N$. If either of the following conditions holds, then learning is perfect for player i :⁵

- (i) $NE_i(\mathbf{g}') \cap NE_i(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$.
- (ii) there exists $j \in N_i(\mathbf{g})$ such that $NE_j(\mathbf{g}') \cap NE_j(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$;

Theorem 2 strengthens behavioral convergence to full identification. An observed action is always informative: it refines a player's beliefs by ruling out networks inconsistent with it. Identification asks more - that this refinement continue until a single network remains. This happens when distinct structures induce distinct actions, so heterogeneity in actions across structures is what pins the network down. The two conditions require this heterogeneity in an action i can observe. Condition (i) asks it of i 's own action, so i pins the structure down from what she plays. Condition (ii) asks it of a neighbor: some $j \in N_i(\mathbf{g})$ acts differently under each network, so watching j pins it down. When actions are homogeneous across structures, refinement stalls short of identification - even though behavior still converges correctly by Theorem 1.

Taken together, Theorems 1 and 2 characterize the limits of decentralized learning in this environment. Local observations are sufficient to ensure correct long-run behavior. Once learning stabilizes, agents act as if the interaction structure were known, even though their

⁵This separation condition is a sufficient identification condition rather than a restriction. Under uniqueness, it reduces to injectivity of the map $\mathbf{g}' \mapsto a_k^c(\mathbf{g}')$. With multiple equilibria, it requires the projected equilibrium sets to be disjoint across structures. In environments with rich action spaces, observed actions may reveal beliefs or informational states; see Arieli and Mueller-Frank (2017) and Molavi et al. (2018). In the linear-quadratic network game (Ballester et al., 2006), equilibrium actions are generically one-to-one in the network, as we show below.

knowledge of the networks may not be complete. However, identification of the interaction structure itself requires a stronger condition, namely that equilibrium behavior should be sufficiently informative to distinguish between observationally equivalent structures. Behavior and knowledge thus need not coincide. Agents may behave optimally relative to the true structure without ever learning it. This distinction reflects a fundamental feature of environments with endogenous and locally generated information showing that learning is constrained not only by what is observed, but also by how equilibrium behavior encodes underlying primitives. In this sense, information along the path of play is determined jointly by the interaction structure and the equilibrium mapping from structure to actions.

4 Quadratic Network Games

We illustrate the general coverage results in a linear-quadratic network game, which provides a transparent interpretation of both convergence and identification. In this environment, equilibrium actions admit a closed-form representation that makes explicit how local interactions aggregate information about the global structure.

4.1 Environment

For each player $i \in N$, let $A_i = \mathbb{R}$ and define stage payoffs by

$$u_i(a, \mathbf{g}) = a_i - \frac{1}{2}a_i^2 + \lambda a_i \sum_{j \in N} g_{ij}(\mathbf{g})a_j. \quad (1)$$

where, $g_{ij}(\mathbf{g}) = 1$ if $j \in N_i(\mathbf{g})$, and 0 otherwise. This is the linear-quadratic network game of [Ballester et al. \(2006\)](#), among the most widely studied models in the networks literature. Player i chooses an action that trades off a quadratic cost against complementarities with the actions of her neighbors. The interaction structure affects payoffs only through the set of neighbors $N_i(\mathbf{g})$, so the local-spillovers property of the general model is preserved.

Define $G(\mathbf{g}) = [g_{ij}(\mathbf{g})]_{1 \leq i, j \leq n}$ as the adjacency matrix representation of the interaction structure. Let $\bar{\rho} := \max_{\mathbf{h} \in \mathcal{G}_n} \rho(G(\mathbf{h}))$, and suppose $0 < \lambda < 1/\bar{\rho}$. Then, for any $\mathbf{g} \in \mathcal{G}_n$, the unique pure Nash equilibrium of the complete-information stage game is given by

$$a^c(\mathbf{g}) = (I - \lambda G(\mathbf{g}))^{-1} \mathbf{1}, \quad (2)$$

which corresponds to the Katz–Bonacich centrality vector of the network $\mathbf{g}$.

We now connect this structure to the general results. Let $\mathbf{g}$ be the realized interaction structure, and suppose learning ends at period τ^* under σ^* . By Theorem 1, equilibrium play converges to the complete-information Nash equilibrium under $\mathbf{g}$. Combining this with (2) yields:

Corollary 1. *Suppose payoffs are given by (1). If learning ends at period τ^* under σ^* and the realized interaction structure is $\mathbf{g}$, then*

$$a_i^\tau(\mathbf{g}, \sigma^*) = \left[(I - \lambda G(\mathbf{g}))^{-1} \mathbf{1} \right]_i \quad \forall i \in N, \forall \tau \geq \tau^*.$$

The corollary shows that even though players initially observe only their local neighborhoods and must infer the rest of the network indirectly, equilibrium play aggregates information across the network. Once learning stabilizes, each player behaves exactly as if the entire interaction structure were known. Any remaining uncertainty is confined to structures that are observationally and behaviorally indistinguishable.

4.2 Identification via Network Walks

The centrality representation also clarifies when equilibrium actions identify the underlying interaction structure. Expanding (2) as a Neumann series, we obtain

$$a^c(\mathbf{g}) = \sum_{k=0}^{\infty} \lambda^k G(\mathbf{g})^k \mathbf{1}. \quad (3)$$

For each player i and $k \geq 0$, define

$$W_i^k(\mathbf{g}) := [G(\mathbf{g})^k \mathbf{1}]_i.$$

Because $G(\mathbf{g})$ is the adjacency matrix, $W_i^k(\mathbf{g})$ counts the number of walks of length k that end at player i .

Thus,

$$a_i^c(\mathbf{g}) = \sum_{k=0}^{\infty} \lambda^k W_i^k(\mathbf{g}),$$

so equilibrium actions encode the entire walk structure of the network around each player.

This is where behavior meets identification. Player i 's action aggregates walks of every length, so it reflects the network's higher-order structure, not just her direct links. Two

networks that differ at any walk length must then differ in her action, for almost every λ ⁶. Behavior separates them, and identification follows.

Proposition 1. *Fix $i \in N$ and two interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$. Suppose there exists $k \geq 1$ such that $W_i^k(\mathbf{g}) \neq W_i^k(\mathbf{g}')$. Then $a_i^c(\mathbf{g}) \neq a_i^c(\mathbf{g}')$ for almost all values of λ satisfying $0 < \lambda < 1/\max\{\rho(G(\mathbf{g})), \rho(G(\mathbf{g}'))\}$.*

The intuition follows directly from (3). Equilibrium actions aggregate walk counts at all lengths. If two networks differ at any walk length, then their corresponding power series generically differ, and hence define distinct equilibrium actions for almost all admissible values of λ .

An additional and novel implication of Proposition 1 is that equilibrium actions in the linear-quadratic network game are generically unique across interaction structures. While the game admits a unique Nash equilibrium for each fixed network, the proposition shows that, for almost all values of λ , interaction structures that differ in the relevant walk profile induce distinct equilibrium action profiles. Thus, equilibrium behavior is generically one-to-one in the underlying network's walk profile. This reflects the fundamental underlying structural property of the model, that is, equilibrium actions encode the network through the pattern of walks. In this sense, the linear-quadratic environment exhibits a strong form of behavioral identifiability giving rise to the following corollary.

Corollary 2. *Fix $\mathbf{g} \in \mathcal{G}_n$ and $i \in N$. Suppose some $j \in \{i\} \cup N_i(\mathbf{g})$ can distinguish every pair of interaction structures in $\mathcal{G}_n$ through its walk profile, i.e., $\mathbf{g}' \neq \mathbf{g}'' \in \mathcal{G}_n$ implies $W_j^k(\mathbf{g}') \neq W_j^k(\mathbf{g}'')$ for some $k \geq 0$. Then, if learning terminates at period τ^* under σ^* , player i achieves perfect learning for almost all admissible values of λ .*

The linear-quadratic environment gives a concrete reading of the general results. Information about the network propagates through walks, and equilibrium actions aggregate these effects via the Katz–Bonacich representation; once learning stabilizes, behavior reflects the entire upstream network relevant for payoffs. This is Theorem 1: local observation and equilibrium response suffice to recover the payoff-relevant structure. Proposition 1 adds that identification turns on whether these effects uniquely encode the network: when walk profiles differ, actions separate the structures for almost all λ , and learning is perfect. Convergence thus reflects how payoff-relevant influence propagates through the network; identification, whether that influence generates distinct walk profiles.

⁶“almost all λ ” means for all admissible λ outside a set of Lebesgue measure zero. In each of our results this exceptional set is finite and it consists of the roots of a polynomial in λ so the property holds for all but finitely many values.

The linear-quadratic utility is used only for the closed-form Katz–Bonacich representation and the walk-based identification result; convergence itself does not require interior equilibria. It applies to any network game with local payoff spillovers, including the linear-best-reply class of [Bramoullé et al. \(2014\)](#), which comprises of quadratic games, local public-good games and R&D networks, with multiplicity in equilibria and corner solutions. In all of these, once learning terminates under a myopic Bayesian equilibrium, the realized profile is a complete-information Nash equilibrium of the realized network. Corner solutions and multiplicity therefore do not break the bridge from learning to complete-information behavior.

5 Speed of Learning

Section 3 established two conclusions. First, for any fixed myopic Bayesian equilibrium, each player’s observational-equivalence class shrinks only through elimination and therefore stabilizes in finite time. Second, once learning stabilizes, equilibrium play coincides with the complete-information Nash equilibrium of the realized interaction structure. We now ask a natural follow up question: *how long can the learning process last?*

Fix an arbitrary myopic Bayesian equilibrium σ^* . By Lemma 1, the earliest period at which learning ends under σ^* is well defined. Denote this time by

$$\tau^*(\sigma^*) := \min \left\{ \tau \geq 1 : [\mathbf{g}]_{i,\tau+1} = [\mathbf{g}]_{i,\tau} \text{ for all } i \in N, \mathbf{g} \in \mathcal{G}_n \right\}.$$

Because the classes $[\mathbf{g}]_{i,\tau}$ are non-increasing in τ , equality between periods τ and $\tau + 1$ is equivalent to stabilization from period τ onward.

Two forces determine $\tau^*(\sigma^*)$. The first is purely combinatorial and uses only the finiteness of $\mathcal{G}_n$. The second reflects the topology of local monitoring: information propagates only along directed paths in the interaction structure, so that learning speed is governed by the depth of these paths.

5.1 A Coarse Bound from Finite Refinements

For each player i and period $\tau \geq 1$, let

$$\Pi_i^\tau(\sigma^*) := \left\{ [\mathbf{g}]_{i,\tau} : \mathbf{g} \in \mathcal{G}_n \right\}, \quad b_i^\tau(\sigma^*) := |\Pi_i^\tau(\sigma^*)|.$$

The collection $\Pi_i^\tau(\sigma^*)$ is the partition of $\mathcal{G}_n$ induced by player i ’s period- τ private histories and provides a representation of player i ’s information at period τ . Each element collects

interaction structures that are observationally indistinguishable given i 's private history. As learning proceeds, this partition becomes finer since new observations refine indistinguishable structures into smaller sets. The number of cells $b_i^\tau(\sigma^*)$ therefore proxies the amount of information player i has acquired. Bounding the speed of learning reduces to bounding how many such refinements can occur before the partition stabilizes.

Because private histories expand over time, for every $i \in N$, $\tau \geq 1$, and $\mathbf{g} \in \mathcal{G}_n$,

$$[\mathbf{g}]_{i,\tau+1} \subseteq [\mathbf{g}]_{i,\tau}.$$

Thus, $\Pi_i^{\tau+1}(\sigma^*)$ is a refinement of $\Pi_i^\tau(\sigma^*)$, and the sequence

$$b_i^1(\sigma^*), b_i^2(\sigma^*), b_i^3(\sigma^*), \dots$$

is weakly increasing.

Theorem 3. *Fix a myopic Bayesian equilibrium σ^* . Suppose that σ^* is partition-stationary in the following sense: whenever $\Pi_i^{\tau+1}(\sigma^*) = \Pi_i^\tau(\sigma^*)$, for every $i \in N$, we have $a_i^{\tau+1}(\mathbf{g}, \sigma^*) = a_i^\tau(\mathbf{g}, \sigma^*)$, for every $i \in N$, and every $\mathbf{g} \in \mathcal{G}_n$. Then, $\tau^*(\sigma^*) \leq 1 + n(|\mathcal{G}_n| - 1)$.*

Theorem 3 follows from a simple counting argument. Each strict refinement of $\Pi_i^\tau(\sigma^*)$ increases $b_i^\tau(\sigma^*)$ by at least one, and for each player i , the number of such increases is bounded by $|\mathcal{G}_n| - 1$. Aggregating across players yields the stated bound.

This bound is coarse and depends only on the size of the candidate class. It does not exploit the structure of local monitoring and treats all refinements as equally informative. Information in our environment, however, propagates through the interaction structure along constrained paths. The next subsection uses this underlying structure to obtain sharper bounds.

The stationarity condition is used only to convert a bound on the number of strict refinements into a bound on calendar time. Without this condition, the finite state space still bounds the number of times information partitions can strictly refine, but it does not bound when those refinements occur. In particular, a nonstationary equilibrium selection could keep the same information partition for many periods and reveal already-held information only at a later calendar date.⁷

⁷For example, suppose player j can distinguish two interaction structures $\mathbf{g}$ and $\mathbf{g}'$, while player i cannot, and i observes j . If two actions are payoff-equivalent for j , then an equilibrium selection may prescribe the same action for j under $\mathbf{g}$ and $\mathbf{g}'$ for T periods and then prescribe different actions in period $T + 1$. Player i 's partition would not refine before period $T + 1$, but would refine afterward.

5.2 A Distance Bound under Action Separation

A sharper bound can be found once we exploit the fact that learning in our model is local. Under local monitoring, player i never observes the whole interaction structure directly. Instead, she learns about distant parts of the network only indirectly through the actions of the players she observes. This induces a propagation structure: information about a remote node must first affect that node's own history, then that node's action, then a downstream player's signal, and so on. Information can thus travel at most one edge per period.

To convert this propagation structure into a sharper bound, however, it is not enough that actions respond to histories. Observed actions must also reveal the histories that generated them. Otherwise, information may be generated upstream but fail to be decoded by downstream players.

Assumption 2. *Fix a myopic Bayesian equilibrium σ^* . For every player $i \in N$, every period $\tau \geq 1$, and every $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$,*

$$[\mathbf{g}]_{i,\tau} \neq [\mathbf{g}']_{i,\tau} \implies a_i^\tau(\mathbf{g}, \sigma^*) \neq a_i^\tau(\mathbf{g}', \sigma^*).$$

Assumption 2 again demands heterogeneity, now requiring a player's action to vary across her informational classes so that distinct classes leave distinct traces in what she does.⁸ Behavior then doesn't merely respond to local information but also transmits that information. So observing a neighbor's action is equivalent to learning what she has inferred so far. Were the condition to fail, a player would act identically in two distinct classes, and an observer could neither tell the classes apart nor recover what separates them.

To formalize the resulting propagation structure further, we orient links as follows

$$j \rightarrow i \iff j \in N_i(\mathbf{g}).$$

Thus, information flows toward the observer.

For a realized interaction structure $\mathbf{g}$, define the directed distance from j to i by

$$\text{dist}_{\mathbf{g}}(j, i) := \min \left\{ \ell \in \mathbb{N}_0 : \exists \text{ a directed path } j \rightarrow \cdots \rightarrow i \text{ of length } \ell \right\},$$

with the convention $\text{dist}_{\mathbf{g}}(j, i) = \infty$ if no such path exists. For $r \geq 0$, define the upstream

⁸Assumption 2 requires equilibrium actions to reveal current informational states along the equilibrium path. This is related to the idea that actions can encode beliefs or private information when the action space is sufficiently rich; see, for example, [Arieli and Mueller-Frank \(2017\)](#).

r -neighborhood of i by

$$N_i^r(\mathbf{g}) := \{j \in N : \text{dist}_{\mathbf{g}}(j, i) \leq r\}.$$

Hence,

$$N_i^0(\mathbf{g}) = \{i\}, \quad N_i^1(\mathbf{g}) = \{i\} \cup N_i(\mathbf{g}).$$

For $r \geq 1$, define the rooted upstream radius- r structure of player i by

$$R_i^r(\mathbf{g}) := (N_j(\mathbf{g}))_{j \in N_i^{r-1}(\mathbf{g})}.$$

This object records the incoming neighborhoods of all nodes whose directed distance to i is at most $r - 1$. Equivalently, it is the subset of the interaction structure that can possibly affect player i 's observations within r rounds of play. Nodes outside this rooted upstream neighborhood are too far away to have influenced what player i has seen by time r .

The next lemma summarizes the informational content of a player's history.

Lemma 2. *Fix a myopic Bayesian equilibrium σ^* . For every $i \in N$, $\tau \geq 1$, and $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$,*

$$R_i^\tau(\mathbf{g}) = R_i^\tau(\mathbf{g}') \Rightarrow h_i^\tau(\mathbf{g}, \sigma^*) = h_i^\tau(\mathbf{g}', \sigma^*).$$

Under Assumption 2, the converse also holds. Hence, under action separation, h_i^τ identifies exactly the rooted upstream radius- τ structure.

Lemma 2 has a simple economic interpretation. By time τ , player i can learn only the part of the network whose consequences could have reached her through $\tau - 1$ rounds of locally observed play. Under Assumption 2, no information is lost, so this bound is exact and player i 's history at time τ is informationally equivalent to knowing the entire upstream portion of the network out to radius τ . In other words, the relevant state variable for time- τ learning is not the whole interaction structure, but the rooted upstream τ -neighborhood.

This equivalence yields a sharper bound on the speed of learning. For every interaction structure $\mathbf{g}$, define its directed diameter by

$$\text{diam}(\mathbf{g}) := \max_{i, j \in N} \{\text{dist}_{\mathbf{g}}(j, i) : \text{dist}_{\mathbf{g}}(j, i) < \infty\},$$

and define the maximal directed diameter across the candidate class by

$$d(\mathcal{G}_n) := \max_{\mathbf{g} \in \mathcal{G}_n} \text{diam}(\mathbf{g}).$$

Proposition 2. *Suppose Assumption 2 holds. Then*

$$\tau^*(\sigma^*) \leq d(\mathcal{G}_n) + 1.$$

Proposition 2 shows that the speed of learning is governed by the topology of information flow. When the candidate class has small diameter, all strategically relevant uncertainty is exhausted quickly because information propagates through the entire network in only a few rounds. When the diameter is large, learning is slower because the information encoded in actions exerted at distant parts of the network must pass through more intermediate neighbors and in turn reflected in those neighbors' actions before they become observable to the agent. The proposition therefore shows that local monitoring creates a propagation friction, while equilibrium separation determines whether that local information can be decoded without loss. Because the bound is governed by diameter rather than by the number of players, learning is rapid whenever the diameter is short, as in small-world networks (Watts and Strogatz, 1998). The number of players nonetheless enters indirectly, since it bounds the diameter itself.

A direct corollary is that the learning time is at most linear in the number of players.

Corollary 3. *Suppose Assumption 2 holds. Then*

$$\tau^*(\sigma^*) \leq n.$$

In summary, the two bounds are complementary. While Theorem 3 is general, resulting from the finite number of possible interaction structures candidate model, it abstracts from the structure of information flow. Proposition 2 exploits this structure to provide sharper bounds, but requires Assumption 2 to ensure that observed behavior fully reveals current informational states.

6 Conclusion

This paper studies repeated games in which agents do not know how observations are structured and must learn the interaction environment from locally observed behavior. We show that uncertainty about monitoring can be represented as uncertainty about an underlying interaction network, allowing the problem to be analyzed within a standard Bayesian framework. Our results establish that equilibrium behavior converges in finite time to the complete-information benchmark even when the interaction structure is not identified, and

characterize when and how identification occurs. The analysis highlights a distinction between behavioral convergence and structural learning and shows that both are governed by the informativeness of the stage-game equilibrium mapping from interaction structures to actions.

More broadly, the framework provides a tractable way to study environments in which the flow of information is itself uncertain. It suggests that many settings typically modeled with imperfect monitoring can be reinterpreted as problems of learning an underlying interaction structure. We have framed interactions as bilateral links, but the analysis uses only that payoffs and observations are governed by a common latent structure from a finite class. The same logic extends to richer structures, such as group or coalitional interactions modeled as hypergraphs, and multilayer relations. The convergence and identification logic therefore extends to such settings, with the speed of learning governed by the corresponding notion of distance. An important direction for future work is to allow for forward-looking behavior and strategic incentives to influence learning, as well as to study environments in which the interaction structure evolves over time.

Proofs

Proof of Lemma 1.

Fix a myopic Bayesian equilibrium σ^* and a player $i \in N$. Let $\Pi_i^\tau(\sigma^*) = \Pi_i^\tau$ be the partition of $\mathcal{G}_n$ induced by the equivalence relation $\sim_{i,\tau}$ under σ^* . Formally, $\Pi_i^\tau = \{[\mathbf{g}]_{i,\tau} : \mathbf{g} \in \mathcal{G}_n\}$. We first show that $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ .

For any $\tau \geq 1$ choose any equivalence class $[\mathbf{g}]_{i,\tau+1} \in \Pi_i^{\tau+1}$ arbitrarily and let $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$. Then, from the definition of histories we get,

$$h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*) \Rightarrow h_i^\tau(\mathbf{g}, \sigma^*) = h_i^\tau(\mathbf{g}', \sigma^*).$$

From the last equality it follows that, $\mathbf{g} \sim_{i,\tau} \mathbf{g}' \Rightarrow \mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Hence, for any $\mathbf{g} \in \mathcal{G}_n$ we have $[\mathbf{g}]_{i,\tau+1} \subseteq [\mathbf{g}]_{i,\tau}$. Therefore, $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ .

Let $b_i^\tau = |\Pi_i^\tau|$. Since $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ , the sequence $(b_i^\tau)_{\tau \geq 1}$ is weakly increasing. Also, because each Π_i^τ is a partition of the finite set $\mathcal{G}_n$, we have $1 \leq b_i^\tau \leq |\mathcal{G}_n|$ for every $\tau \geq 1$. Hence $(b_i^\tau)_{\tau \geq 1}$ is an integer-valued weakly increasing sequence that is bounded above. Therefore, there exists $T_i < \infty$ such that $b_i^\tau = b_i^{T_i}$ for all $\tau \geq T_i$.

Fix any $\tau \geq T_i$. Since $\Pi_i^{\tau+1}$ refines Π_i^τ and the two partitions have the same number of elements, this refinement cannot be strict. Thus $\Pi_i^{\tau+1} = \Pi_i^\tau$. Repeating the same argument period by period, we obtain $\Pi_i^\tau = \Pi_i^{T_i}$ for every $\tau \geq T_i$. Thus the equivalence class in the partition containing $\mathbf{g}$ is the same at periods τ and T_i . Therefore, $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,T_i}$ for every $\mathbf{g} \in \mathcal{G}_n$ and every $\tau \geq T_i$.

Since N is finite, let $T = \max_{i \in N} T_i$. Then $T < \infty$, and for every player i , every interaction structure $\mathbf{g} \in \mathcal{G}_n$, and every period $\tau \geq T$, we have $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,T}$. Thus no player's observational-equivalence class changes after period T . Therefore, learning ends in finite time. $\blacksquare$

Proof of Theorem 1.

Fix the myopic Bayesian equilibrium σ^* , a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$, a period $\tau \geq \tau^*$, and a player $i \in N$. Write h_i^τ for $h_i^\tau(\mathbf{g}, \sigma^*)$. Since $h_i^\tau = h_i^\tau(\mathbf{g}, \sigma^*)$ is generated by the realized interaction structure $\mathbf{g}$ under σ^* , and since the prior has full support, we have $P_{i,\sigma^*}(h_i^\tau) \geq p(\mathbf{g}) > 0$. Hence $h_i^\tau \in P_i(\sigma^*)$, so the posterior at h_i^τ is determined by Bayes' rule.

By Bayes' rule, player i assigns positive probability at h_i^τ exactly to those interaction structures that generate the same period- τ private history as $\mathbf{g}$. Thus $\mu_{i,\sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) > 0$ if and only if $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Moreover, because the first component of player i 's private history is her observed neighborhood, every $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$ satisfies $N_i(\mathbf{g}') = N_i(\mathbf{g})$.

Now take any $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Since learning has ended at τ^* and $\tau \geq \tau^*$, the equivalence class of $\mathbf{g}$ for player i is the same at periods τ and $\tau + 1$. Hence $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$ as well. Therefore $h_i^{\tau+1}(\mathbf{g}', \sigma^*) = h_i^{\tau+1}(\mathbf{g}, \sigma^*)$. Since a period- $(\tau + 1)$ history is obtained from the period- τ history by adding the period- τ signal, it follows that $z_i^\tau(\mathbf{g}', \sigma^*) = z_i^\tau(\mathbf{g}, \sigma^*)$. Together with $N_i(\mathbf{g}') = N_i(\mathbf{g})$, equality of signals implies that the actions observed by player i are the same under $\mathbf{g}'$ and under $\mathbf{g}$; that is, $a_{N_i(\mathbf{g}')}^\tau(\mathbf{g}', \sigma^*) = a_{N_i(\mathbf{g})}^\tau(\mathbf{g}, \sigma^*)$.

We can now compare player i 's payoff across all interaction structures in the support of her posterior. Fix any $a_i \in A_i$ and any $\mathbf{g}'$ such that $\mu_{i,\sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) > 0$. From the previous paragraph, $\mathbf{g}'$ has the same neighborhood for player i as $\mathbf{g}$, and the corresponding local action vector is also the same. Therefore, by the local payoff-equivalence assumption,

$$u_i\left(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}', \sigma^*)); \mathbf{g}'\right) = u_i\left(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g}\right).$$

Thus every payoff term that receives positive posterior probability in player i 's myopic objective is identical to the payoff she would obtain under the realized interaction structure

$\mathbf{g}$.

Since the remaining interaction structures have zero posterior probability, player i 's posterior expected payoff at h_i^τ is the same function of a_i as the complete-information payoff under the realized interaction structure $\mathbf{g}$, i.e.

$$\sum_{\mathbf{g}' \in \mathcal{G}_n} \mu_{i, \sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) u_i(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}', \sigma^*)); \mathbf{g}') = u_i(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g})$$

for every $a_i \in A_i$.

Therefore, by the myopic Bayesian equilibrium condition, $\sigma_i^{*\tau}(h_i^\tau)$ maximizes $u_i(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g})$ over A_i . Using the definition of induced play, this gives

$$a_i^\tau(\mathbf{g}, \sigma^*) \in \arg \max_{a_i \in A_i} u_i(a_i, a_{-i}^\tau(\mathbf{g}, \sigma^*); \mathbf{g}).$$

Hence $a_i^\tau(\mathbf{g}, \sigma^*)$ is a best response to $a_{-i}^\tau(\mathbf{g}, \sigma^*)$ in the complete-information stage game induced by $\mathbf{g}$.

Since this argument applies to every player $i \in N$, the realized action profile $a^\tau(\mathbf{g}, \sigma^*)$ is a Nash equilibrium of the complete-information stage game under $\mathbf{g}$. Therefore, for every $\tau \geq \tau^*$,

$$a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g}),$$

where $NE(\mathbf{g})$ denotes the set of pure Nash equilibria of the complete-information stage game induced by $\mathbf{g}$. ■

Proof of Theorem 2.

Fix the myopic Bayesian equilibrium σ^* and a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$. Suppose learning ends at τ^* under σ^* .

First suppose there exists $j \in N_i(\mathbf{g})$ such that $NE_j(\mathbf{g}') \cap NE_j(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$. If possible, let us assume that learning is not perfect for player i . Then there exists $\mathbf{g}' \neq \mathbf{g}$ such that $\mathbf{g}' \in [\mathbf{g}]_{i, \tau^*}$.

Since learning ends at τ^* , the equivalence class of $\mathbf{g}$ for player i is the same at periods τ^* and $\tau^* + 1$. Hence $\mathbf{g}' \in [\mathbf{g}]_{i, \tau^*+1}$ as well. Therefore $h_i^{\tau^*+1}(\mathbf{g}', \sigma^*) = h_i^{\tau^*+1}(\mathbf{g}, \sigma^*)$, and so the period- τ^* signals of player i coincide under $\mathbf{g}'$ and $\mathbf{g}$. Moreover, because $\mathbf{g}' \in [\mathbf{g}]_{i, \tau^*}$, the initial component of player i 's history gives $N_i(\mathbf{g}') = N_i(\mathbf{g})$. Since $j \in N_i(\mathbf{g})$, equality of signals implies $a_j^{\tau^*}(\mathbf{g}', \sigma^*) = a_j^{\tau^*}(\mathbf{g}, \sigma^*)$.

By Theorem 1, applied to both interaction structures, $a^{\tau^*}(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ and $a^{\tau^*}(\mathbf{g}', \sigma^*) \in NE(\mathbf{g}')$. Hence,

$$a_j^{\tau^*}(\mathbf{g}, \sigma^*) = a_j^{\tau^*}(\mathbf{g}', \sigma^*) \in NE_j(\mathbf{g}) \cap NE_j(\mathbf{g}').$$

This contradicts the assumed separation of $NE_j(\cdot)$ across distinct interaction structures. Therefore $[\mathbf{g}]_{i, \tau^*} = \{\mathbf{g}\}$.

Now suppose instead that $NE_i(\mathbf{g}') \cap NE_i(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$. Again if possible, let us assume that learning is not perfect for player i , and choose $\mathbf{g}' \neq \mathbf{g}$ with $\mathbf{g}' \in [\mathbf{g}]_{i, \tau^*}$. Then $h_i^{\tau^*}(\mathbf{g}', \sigma^*) = h_i^{\tau^*}(\mathbf{g}, \sigma^*)$. Since σ_i^* is a pure strategy, equal private histories imply $a_i^{\tau^*}(\mathbf{g}', \sigma^*) = a_i^{\tau^*}(\mathbf{g}, \sigma^*)$.

By Theorem 1, $a^{\tau^*}(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ and $a^{\tau^*}(\mathbf{g}', \sigma^*) \in NE(\mathbf{g}')$. Hence,

$$a_i^{\tau^*}(\mathbf{g}', \sigma^*) = a_i^{\tau^*}(\mathbf{g}, \sigma^*) \in NE_i(\mathbf{g}) \cap NE_i(\mathbf{g}'),$$

contradicting the assumed separation of $NE_i(\cdot)$ across distinct interaction structures. Thus $[\mathbf{g}]_{i, \tau^*} = \{\mathbf{g}\}$. ■

Proof of Corollary 1.

Fix a myopic Bayesian equilibrium σ^* and a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$. Suppose learning ends at period τ^* under σ^* .

By Theorem 1, for every $\tau \geq \tau^*$, $a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$. It remains to characterize the complete-information Nash equilibrium set for the linear-quadratic stage game.

Under (1), player i 's payoff is strictly concave in a_i , and hence the first-order condition is necessary and sufficient. Thus any complete-information Nash equilibrium a under $\mathbf{g}$ satisfies $1 - a_i + \lambda \sum_{j \in N} g_{ij}(\mathbf{g}) a_j = 0$ for every $i \in N$. Stacking these equations gives

$$(I - \lambda G(\mathbf{g}))a = \mathbf{1}.$$

Since $0 < \lambda < 1/\bar{\rho}$ and $\rho(G(\mathbf{g})) \leq \bar{\rho}$, the matrix $I - \lambda G(\mathbf{g})$ is invertible. Therefore the complete-information Nash equilibrium under $\mathbf{g}$ is unique and is given by

$$a^c(\mathbf{g}) = (I - \lambda G(\mathbf{g}))^{-1} \mathbf{1}.$$

Since $a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ for every $\tau \geq \tau^*$, uniqueness implies

$$a_i^\tau(\mathbf{g}, \sigma^*) = \left[(I - \lambda G(\mathbf{g}))^{-1} \mathbf{1} \right]_i \quad \text{for every } i \in N \text{ and every } \tau \geq \tau^*.$$

■

Proof of Proposition 1.

Fix $i \in N$ and two interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$. Write $G := G(\mathbf{g})$ and $G' := G(\mathbf{g}')$, and let $\rho^* := \max\{\rho(G), \rho(G')\}$. For $\lambda \in (0, 1/\rho^*)$, define

$$f_i(\lambda) := e_i^\top (I - \lambda G)^{-1} \mathbf{1} - e_i^\top (I - \lambda G')^{-1} \mathbf{1}.$$

Thus $f_i(\lambda) = a_i^c(\mathbf{g}) - a_i^c(\mathbf{g}')$ whenever the two complete-information equilibrium vectors are well defined.

Since $\lambda\rho(G) < 1$ and $\lambda\rho(G') < 1$ for every $\lambda \in (0, 1/\rho^*)$, both Neumann-series expansions are valid. Hence

$$f_i(\lambda) = \sum_{m=0}^{\infty} \lambda^m [(G^m \mathbf{1})_i - ((G')^m \mathbf{1})_i] = \sum_{m=0}^{\infty} \lambda^m [W_i^m(\mathbf{g}) - W_i^m(\mathbf{g}')].$$

By assumption, there exists $k \geq 1$ such that $W_i^k(\mathbf{g}) \neq W_i^k(\mathbf{g}')$. Therefore the above power series has a nonzero coefficient, so f_i is not identically zero on $(0, 1/\rho^*)$.

It remains to rule out infinitely many coincidences. Each coordinate of $(I - \lambda G)^{-1} \mathbf{1}$ is a rational function of λ , since

$$e_i^\top (I - \lambda G)^{-1} \mathbf{1} = \frac{P_i(\lambda)}{\det(I - \lambda G)}$$

for some polynomial P_i , and similarly for G' . Thus f_i can be written as a rational function $N_i(\lambda)/D_i(\lambda)$, where $D_i(\lambda) \neq 0$ for all $\lambda \in (0, 1/\rho^*)$. Since f_i is not identically zero, N_i is not the zero polynomial. Hence N_i has only finitely many roots.

Therefore there exists a finite and hence null set, $\Lambda_i(\mathbf{g}, \mathbf{g}') \subset (0, 1/\rho^*)$ such that $f_i(\lambda) \neq 0$ for every $\lambda \in (0, 1/\rho^*) \setminus \Lambda_i(\mathbf{g}, \mathbf{g}')$. Equivalently,

$$a_i^c(\mathbf{g}) \neq a_i^c(\mathbf{g}') \quad \text{for all } \lambda \in (0, 1/\rho^*) \setminus \Lambda_i(\mathbf{g}, \mathbf{g}').$$

■

Proof of Corollary 2.

Fix $\mathbf{g} \in \mathcal{G}_n$ and $i \in N$. Suppose there exists $j \in \{i\} \cup N_i(\mathbf{g})$ such that the walk profile $\mathbf{h} \mapsto (W_j^k(\mathbf{h}))_{k \geq 0}$ distinguishes every pair of interaction structures in $\mathcal{G}_n$.

Take any two distinct interaction structures $\mathbf{h}, \mathbf{h}' \in \mathcal{G}_n$. By assumption, there exists $k \geq 0$ such that $W_j^k(\mathbf{h}) \neq W_j^k(\mathbf{h}')$. Since $W_j^0(\mathbf{h}) = W_j^0(\mathbf{h}') = 1$, this difference must occur at some $k \geq 1$. Proposition 1 then implies that there is a finite and hence null set, $\Lambda_j(\mathbf{h}, \mathbf{h}') \subset (0, 1/\bar{\rho})$ such that $a_j^\epsilon(\mathbf{h}) \neq a_j^\epsilon(\mathbf{h}')$ for every $\lambda \in (0, 1/\bar{\rho}) \setminus \Lambda_j(\mathbf{h}, \mathbf{h}')$.

Since $\mathcal{G}_n$ is finite, the union $\Lambda_j := \bigcup_{\mathbf{h} \neq \mathbf{h}'} \Lambda_j(\mathbf{h}, \mathbf{h}')$ is finite. Hence, for every $\lambda \in (0, 1/\bar{\rho}) \setminus \Lambda_j$, the map $\mathbf{h} \mapsto a_j^\epsilon(\mathbf{h})$ is injective on $\mathcal{G}_n$.

In the linear-quadratic stage game, the complete-information Nash equilibrium under each $\mathbf{h}$ is unique. Therefore $NE_j(\mathbf{h}) = \{a_j^\epsilon(\mathbf{h})\}$ for every $\mathbf{h} \in \mathcal{G}_n$. Thus, for every admissible $\lambda \notin \Lambda_j$, the equilibrium sets $NE_j(\mathbf{h})$ are pairwise disjoint across distinct interaction structures.

If $j = i$, condition (ii) of Theorem 2 applies. If $j \in N_i(\mathbf{g})$, condition (i) of Theorem 2 applies. Therefore, whenever learning terminates at τ^* under σ^* , player i learns the realized interaction structure perfectly: $[\mathbf{g}]_{i, \tau^*} = \{\mathbf{g}\}$. $\blacksquare$

Proof of Theorem 3.

Write $\Pi_i^\tau = \Pi_i^\tau(\sigma^*)$ and $b_i^\tau = b_i^\tau(\sigma^*)$. Fix a player $i \in N$. Since Π_i^τ is a partition of the finite set $\mathcal{G}_n$, we have $1 \leq b_i^\tau \leq |\mathcal{G}_n|$ for every $\tau \geq 1$. Moreover, since $\Pi_i^{\tau+1}$ refines Π_i^τ , the sequence $(b_i^\tau)_{\tau \geq 1}$ is weakly increasing. Hence, for each player i , this sequence can strictly increase at most $|\mathcal{G}_n| - 1$ times.

We now show that, before learning ends, some player's partition must strictly refine in every period. Fix any period τ . Suppose, toward a contradiction, that no player's partition strictly refines from τ to $\tau + 1$. Since each $\Pi_i^{\tau+1}$ refines Π_i^τ , this means $\Pi_i^{\tau+1} = \Pi_i^\tau$, for every $i \in N$. By the partition-stationarity condition,

$$a_i^{\tau+1}(\mathbf{g}, \sigma^*) = a_i^\tau(\mathbf{g}, \sigma^*) \quad \text{for every } i \in N \text{ and every } \mathbf{g} \in \mathcal{G}_n.$$

Therefore, the period- $(\tau + 1)$ action profile carries exactly the same graph-dependent information as the period- τ action profile. Since the period- τ action profile did not refine any player's partition, the period- $(\tau + 1)$ action profile cannot refine any player's partition either. Hence $\Pi_i^{\tau+2} = \Pi_i^{\tau+1}$, for every $i \in N$. Repeating the same argument inductively gives

$$\Pi_i^s = \Pi_i^\tau \quad \text{for every } s \geq \tau \text{ and every } i \in N.$$

Thus learning has already stabilized at period τ .

Therefore, if $\tau < \tau^*(\sigma^*)$, it cannot be the case that all players' partitions remain unchanged from τ to $\tau + 1$. Hence there must exist some player $i \in N$ and some interaction structure $\mathbf{g} \in \mathcal{G}_n$ such that $[\mathbf{g}]_{i,\tau+1} \subset [\mathbf{g}]_{i,\tau}$. Since $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ , this strict containment means that the refinement is strict, and therefore $b_i^{\tau+1} > b_i^\tau$.

Thus every period before $\tau^*(\sigma^*)$ generates at least one strict increase in one of the n sequences $(b_i^\tau)_{\tau \geq 1}$, $i \in N$. Since each such sequence can strictly increase at most $|\mathcal{G}_n| - 1$ times, the total number of strict increases across all players is at most $n(|\mathcal{G}_n| - 1)$. Hence the number of periods before $\tau^*(\sigma^*)$ is at most $n(|\mathcal{G}_n| - 1)$, and so

$$\tau^*(\sigma^*) \leq 1 + n(|\mathcal{G}_n| - 1).$$

■

Proof of Lemma 2.

We prove the two implications by induction on τ .

For $\tau = 1$, the claim follows directly from the definitions. Indeed, $h_i^1(\mathbf{g}, \sigma^*) = N_i(\mathbf{g})$, while $R_i^1(\mathbf{g})$ records exactly $N_i(\mathbf{g})$ since $N_i^0(\mathbf{g}) = \{i\}$. Hence $R_i^1(\mathbf{g}) = R_i^1(\mathbf{g}')$ if and only if $h_i^1(\mathbf{g}, \sigma^*) = h_i^1(\mathbf{g}', \sigma^*)$. Thus both implications hold at $\tau = 1$.

Fix $\tau \geq 1$ and suppose the result holds at period τ for every player. We first prove the forward implication at period $\tau + 1$. Suppose $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$. Then, in particular, $N_i(\mathbf{g}) = N_i(\mathbf{g}')$. For every $j \in N_i(\mathbf{g}) \cup \{i\}$ and every $\ell \in N_j^{\tau-1}(\mathbf{g})$,

$$\text{dist}_{\mathbf{g}}(\ell, i) \leq \text{dist}_{\mathbf{g}}(\ell, j) + \text{dist}_{\mathbf{g}}(j, i) \leq \tau.$$

Hence $N_j^{\tau-1}(\mathbf{g}) \subseteq N_i^\tau(\mathbf{g})$. Since

$$R_i^{\tau+1}(\mathbf{g}) = (N_\ell(\mathbf{g}))_{\ell \in N_i^\tau(\mathbf{g})} \quad \text{and} \quad R_j^\tau(\mathbf{g}) = (N_\ell(\mathbf{g}))_{\ell \in N_j^{\tau-1}(\mathbf{g})},$$

the latter is determined by the former. Therefore $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$ implies $R_j^\tau(\mathbf{g}) = R_j^\tau(\mathbf{g}')$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. By the induction hypothesis, $h_j^\tau(\mathbf{g}, \sigma^*) = h_j^\tau(\mathbf{g}', \sigma^*)$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. Since strategies are pure, these equal histories imply equal period- τ actions for all those players. Together with $N_i(\mathbf{g}) = N_i(\mathbf{g}')$, this gives $z_i^\tau(\mathbf{g}, \sigma^*) = z_i^\tau(\mathbf{g}', \sigma^*)$. Since the period- τ histories of player i are also equal by the induction hypothesis, the recursive definition of histories gives $h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*)$.

We now prove the converse at period $\tau + 1$ under Assumption 2. Suppose $h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*)$. Then both the period- τ histories and the period- τ signals of player i coincide under $\mathbf{g}$ and $\mathbf{g}'$. In particular, $N_i(\mathbf{g}) = N_i(\mathbf{g}')$, and equality of signals gives $a_j^\tau(\mathbf{g}, \sigma^*) = a_j^\tau(\mathbf{g}', \sigma^*)$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$.

By the contrapositive of Assumption 2, these equal actions imply $[\mathbf{g}]_{j,\tau} = [\mathbf{g}']_{j,\tau}$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. Hence $h_j^\tau(\mathbf{g}, \sigma^*) = h_j^\tau(\mathbf{g}', \sigma^*)$ for every such j . Applying the induction hypothesis for the converse implication, we obtain $R_j^\tau(\mathbf{g}) = R_j^\tau(\mathbf{g}')$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$.

Finally, $R_i^{\tau+1}(\mathbf{g})$ is determined by $N_i(\mathbf{g})$ together with the collection $\{R_j^\tau(\mathbf{g}) : j \in N_i(\mathbf{g}) \cup \{i\}\}$. Since each of these objects is the same under $\mathbf{g}$ and $\mathbf{g}'$, it follows that $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$.

By induction, the result holds for every $\tau \geq 1$. ■

Proof of Proposition 2.

Suppose Assumption 2 holds and let $D := d(\mathcal{G}_n)$. Fix $i \in N$, $\mathbf{g} \in \mathcal{G}_n$, and $\tau \geq D + 1$. We show that $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,D+1}$.

Take any $\mathbf{g}' \in [\mathbf{g}]_{i,D+1}$. Then $h_i^{D+1}(\mathbf{g}', \sigma^*) = h_i^{D+1}(\mathbf{g}, \sigma^*)$. By Lemma 2 and Assumption 2, this implies $R_i^{D+1}(\mathbf{g}') = R_i^{D+1}(\mathbf{g})$.

Now fix an arbitrary $\tilde{\mathbf{g}} \in \mathcal{G}_n$. By definition of D , every node that reaches i under $\tilde{\mathbf{g}}$ does so along a path of length at most D . Hence $N_i^r(\tilde{\mathbf{g}}) = N_i^D(\tilde{\mathbf{g}})$ for every $r \geq D$. Therefore, for every $\tau \geq D + 1$,

$$R_i^\tau(\tilde{\mathbf{g}}) = (N_j(\tilde{\mathbf{g}}))_{j \in N_i^{\tau-1}(\tilde{\mathbf{g}})} = (N_j(\tilde{\mathbf{g}}))_{j \in N_i^D(\tilde{\mathbf{g}})} = R_i^{D+1}(\tilde{\mathbf{g}}).$$

In particular,

$$R_i^\tau(\mathbf{g}') = R_i^{D+1}(\mathbf{g}') = R_i^{D+1}(\mathbf{g}) = R_i^\tau(\mathbf{g}).$$

Lemma 2 then gives $h_i^\tau(\mathbf{g}', \sigma^*) = h_i^\tau(\mathbf{g}, \sigma^*)$, so $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Hence $[\mathbf{g}]_{i,D+1} \subseteq [\mathbf{g}]_{i,\tau}$.

The reverse inclusion follows from the fact that histories are nested over time: if two structures generate the same period- τ history for player i , then they also generate the same period- $(D+1)$ history. Thus $[\mathbf{g}]_{i,\tau} \subseteq [\mathbf{g}]_{i,D+1}$. Therefore $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,D+1}$ for every $\tau \geq D + 1$.

Since the argument is uniform over i and $\mathbf{g}$, learning stabilizes by period $D + 1$. Hence

$$\tau^*(\sigma^*) \leq D + 1 = d(\mathcal{G}_n) + 1. \quad \blacksquare$$

Proof of Corollary 3.

Fix any $\mathbf{g} \in \mathcal{G}_n$ and any players i, j with

$$\text{dist}_{\mathbf{g}}(j, i) < \infty.$$

By definition, $\text{dist}_{\mathbf{g}}(j, i)$ is the length of a shortest directed path from j to i . Any shortest directed path is simple, since otherwise one could delete a directed cycle and obtain a shorter directed path from j to i . A simple directed path on n players visits each player at most once, so its length is at most $n - 1$. Thus

$$\text{dist}_{\mathbf{g}}(j, i) \leq n - 1.$$

Hence

$$d(\mathcal{G}_n) = \max_{\mathbf{g} \in \mathcal{G}_n} \text{diam}(\mathbf{g}) \leq n - 1.$$

Applying Proposition 2, we obtain

$$\tau^*(\sigma^*) \leq d(\mathcal{G}_n) + 1 \leq n.$$

■

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